

A further conjecture on OEIS A220198, settled by the proof already in hand

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Abstract

OEIS A220198 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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1 The premise

The entry's first terms are 4, 10, 38, 103, 245, 547, 1168, 2395, ..., with offset 1. The sequence satisfies

$$a(n) = 9a(n-1) - 36a(n-2) + 84a(n-3) - 126a(n-4) + 126a(n-5) - 84a(n-6) + 36a(n-7) - 9a(n-8) + a(n-9)$$

a linear recurrence of order 9 with characteristic polynomial

$$q(x) = x^9 - 9x^8 + 36x^7 - 84x^6 + 126x^5 - 126x^4 + 84x^3 - 36x^2 + 9x - 1.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least 9 + 2 further terms after that index.

2 The further conjecture

Separately from the conjecture already settled, the entry records

$$\text{Empirical: } a(n) = (1/20160)*n^8 - (1/5040)*n^7 - (1/480)*n^6 + (107/720)*n^5 - (1151/960)*n^4 + (278/45)*n^3 - (54427/5040)*n^2 + (5323/420)*n - 7 \text{ for } n > 1.$$

The further claim is a CLOSED FORM, and for that the divisibility is not the whole argument. It gives that a and the closed form f satisfy the same linear recurrence; it does not say they are the same solution of it. Write p for the annihilator of f ,

$$p(x) = x^9 - 9x^8 + 36x^7 - 84x^6 + 126x^5 - 126x^4 + 84x^3 - 36x^2 + 9x - 1,$$

of order $D = 9$. Since $q \mid p$ and a satisfies the recurrence of q , a satisfies that of p as well; f satisfies it by construction. Two solutions of a linear recurrence of order D with nonzero leading coefficient agree everywhere as soon as they agree at D consecutive indices, and a and f were checked to agree at $n = 11, \dots, 19$, which lies inside the terms the entry publishes and past the index from which a provably satisfies p . Hence $a(n) = f(n)$ for every n in the claimed range.

Theorem 1. *The conjecture displayed above is true.*

3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly q says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.